$k_B = 1.3806503 \times 10^{-5} pN \cdot \mu m/K$

$T = 273.15 + 37 = 310.15\ K$

$\gamma_N = 6\pi\eta R = 0.0425\ pN \cdot sec/\mu m\ (R = 2.25\ \mu m)$

$\Delta t = 1/1606\ \ sec$

$4k_B T = 0.017128\ pN \cdot \mu m$

$variancef = 4k_B T\left(\frac{\gamma_N}{\Delta t}\right) = (0.02698\ pN)^2$

$variancex = 4k_B T\left(\frac{\Delta t}{\gamma_N}\right) = (0.01589\ \mu m)^2$

$\sqrt{4k_B T\left(\frac{\Delta t}{\gamma_N}\right)} = 0.01589\ \mu m$

$g_{cov}^2 \approx \left\{2 + \frac{1}{(1-\lambda_c^2)}\right\} 4k_B T\left(\frac{\Delta t}{\gamma_N}\right) = \left\{2 + \frac{1}{(1-\lambda_c^2)}\right\}(0.01589\ \mu m)^2$

$g_{cov} \approx \sqrt{2 + \frac{1}{(1-\lambda_c^2)}}\ 0.01589\ \mu m$

$\%\ g_{cov}^2 \approx \left\{2 + \frac{1}{(1-\lambda_c^2)}\right\} 4k_B T\left(\frac{\Delta t}{\gamma_N}\right) = \left\{\frac{1-\lambda_c}{1+\lambda_c}\right\}(0.01589\ \mu m)^2$

$\%\ g_{cov} \approx \sqrt{\frac{1-\lambda_c}{1+\lambda_c}}\ 0.01589\ \mu m$

==========

$a_{pd} = 0.1$; $a_{prd} = 0.1$; $a_{cov} = 0.1$;

$\varepsilon = 0.01$; $\lambda_c = 0.7$; $g_{cov} = 0.01589\ \times \sqrt{2 + \frac{1}{(1-\lambda_c^2)}}$; $g_{cov}^2 = g_{cov} \cdot g_{cov}$;

$\mathrm{Vk} = \mathrm{VkT} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$; $\mathrm{uk} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$; $\mathrm{vk} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$; $\mathrm{wk} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$;

$\mathrm{thetaek} = \begin{bmatrix} \mathrm{thetaxek} \\ \mathrm{thetayek} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$;$d\mathrm{thetaek} = \begin{bmatrix} d\mathrm{thetaxek} \\ d\mathrm{thetayek} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$; $\mathrm{thetamk} = \begin{bmatrix} \mathrm{thetaxmk} \\ \mathrm{thetaymk} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$; $\mathrm{lamdaek} = \begin{bmatrix} \mathrm{lamdaTek} \\ \mathrm{lamdaNek} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $d\mathrm{lamdaek} = \begin{bmatrix} d\mathrm{lamdaTek} \\ d\mathrm{lamdaNek} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$; $\mathrm{lamdamk} = \begin{bmatrix} \mathrm{lamdaTmk} \\ \mathrm{lamdaNmk} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$;

$\mathrm{VDek} = \begin{bmatrix} duek \\ dvek \\ dwek \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$; $\mathrm{VdDek} = \begin{bmatrix} dduek \\ ddvek \\ ddwek \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$;

$$\text{Vdp3ek} = \begin{bmatrix}0\\0\\0\end{bmatrix};\ \text{Vdp2ek} = \begin{bmatrix}0\\0\\0\end{bmatrix};\text{Vdp1ek} = \begin{bmatrix}0\\0\\0\end{bmatrix};$$

$$\mathbf{e}\text{pk} = \begin{bmatrix}0\\0\\0\end{bmatrix};\ \mathbf{e}\text{lamdak} = \begin{bmatrix}0\\0\end{bmatrix};\ \mathbf{e}\text{thetak} = \begin{bmatrix}0\\0\end{bmatrix};$$

$$\mathbf{H} = \begin{bmatrix}\mathbf{I}_{3x3} & 0_{3x3} & 0_{3x3} & 0_{3x3} & 0_{3x3} & 0_{3x2} & 0_{3x2} & 0_{3x2} & 0_{3x2}\\ 0_{2x3} & 0_{2x3} & 0_{2x3} & 0_{2x3} & 0_{2x3} & \mathbf{I}_{2x2} & 0_{2x2} & 0_{2x2} & 0_{2x2}\\ 0_{2x3} & 0_{2x3} & 0_{2x3} & 0_{2x3} & 0_{2x3} & 0_{2x2} & 0_{2x2} & \mathbf{I}_{2x2} & 0_{2x2}\end{bmatrix}$$

$$\text{pdk1} = \begin{bmatrix}0\\0\\0\end{bmatrix};\ \text{pdk2} = \begin{bmatrix}0\\0\\0\end{bmatrix};\ \text{dpmdk1} = \begin{bmatrix}0\\0\\0\end{bmatrix};\ \text{dpmrdk1} = \begin{bmatrix}0\\0\\0\end{bmatrix};$$

$$\mathbf{P}^f = [\quad];$$

$$\%\ \text{Rek1} = g_{cov}^2 \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix};$$

Input: $\mathbf{p}_d[k] = \begin{bmatrix}x_d[k]\\y_d[k]\\z_d[k]\end{bmatrix}$ $\delta\mathbf{p}_d[k] = \begin{bmatrix}\delta x_d[k]\\\delta y_d[k]\\\delta z_d[k]\end{bmatrix}$

Measurement: $\mathbf{p}_m[k]$ Acquired from ADC

% **Control objectives**: ${}^V\delta\mathbf{p}[k+1] = \lambda_c \cdot {}^V\delta\mathbf{p}[k]$

Real-time computation

$$\begin{bmatrix}\delta\hat{u}_d[k]\\\delta\hat{v}_d[k]\\\delta\hat{w}_d[k]\end{bmatrix} = \text{VkT} \cdot \begin{bmatrix}\delta x_d[k]\\\delta y_d[k]\\\delta z_d[k]\end{bmatrix}$$

$$\begin{bmatrix}\delta\hat{u}[k]\\\delta\hat{v}[k]\\\delta\hat{w}[k]\end{bmatrix} = \text{Vdp3ek};\ \begin{bmatrix}duek\\dvek\\dwek\end{bmatrix} = \text{VDek}$$

$$\Delta u[k] = \frac{1}{\text{lamdaTek}}\{\delta\hat{u}_d[k] + (1-\lambda_c)\delta\hat{u}[k] - duek\}$$

$$\Delta v[k] = \frac{1}{\text{lamdaTek}}\{\delta\hat{v}_d[k] + (1-\lambda_c)\delta\hat{v}[k] - dvek\}$$

$$\Delta w[k] = \frac{1}{\text{lamdaNek}}\{\delta\hat{w}_d[k] + (1-\lambda_c)\delta\hat{w}[k] - dwek\}$$

$$f_u[k] = \left(\frac{\gamma_N}{\Delta t}\right)\Delta u[k]$$

$$f_v[k] = \left(\frac{\gamma_N}{\Delta t}\right)\Delta v[k]$$

$$f_w[k] = \left(\frac{\gamma_N}{\Delta t}\right)\Delta w[k]$$

% 3-D position error measurement

If $k = 1$, pdk2 $= \mathbf{p}_d[k]$; pdk1 $= \mathbf{p}_d[k]$;

$\delta\mathbf{p}_m[k] =$ pdk2 $- \mathbf{p}_m[k]$;

If $k = 2$, pdk2 $= \mathbf{p}_d[k-1]$; pdk1 $= \mathbf{p}_d[k-1]$;

$\delta\mathbf{p}_m[k] =$ pdk2 $- \mathbf{p}_m[k]$;

Otherwise, pdk2 $= \mathbf{p}_d[k-2]$; pdk1 $= \mathbf{p}_d[k-1]$;

$\delta\mathbf{p}_m[k] =$ pdk2 $- \mathbf{p}_m[k]$; pdk2 $=$ pdk1; pdk1 $= \mathbf{p}_d[k]$;

$$^{V}\delta\mathbf{p}_m[k] = \begin{bmatrix}\delta u_m[k]\\ \delta v_m[k]\\ \delta w_m[k]\end{bmatrix} = \text{VkT}\ \delta\mathbf{p}_m[k]$$

$\delta\mathbf{p}_{md}[k] = \left(1 - a_{pd}\right)\cdot$ dpmdk1 $+ a_{pd}\delta\mathbf{p}_m[k]$**;** (Deterministic component)

$\delta\mathbf{p}_{mr}[k] = \delta\mathbf{p}_m[k] - \delta\mathbf{p}_{md}[k]$**;** (Random component)

$\delta\mathbf{p}_{mrd}[k] = \left(1 - a_{prd}\right)\cdot$ dpmrdk1 $+ a_{prd}\delta\mathbf{p}_{mr}[k]$

$\delta\mathbf{p}_{mrr}[k] = \delta\mathbf{p}_{mr}[k] - \delta\mathbf{p}_{mrd}[k]$**;**

% $\hat{\mathbf{R}}[k] = (1 - a_{cov})\hat{\mathbf{R}}[k-1] + a_{cov}\delta\mathbf{p}_{mrr}[k]\delta\mathbf{p}_{mrr}^T[k]$

% **Real-time measurement of $\boldsymbol{\Lambda}(\lambda_\parallel, \lambda_\perp)$** and $\mathbf{V}\left(\theta_x, \theta_y\right)$:

$\delta\mathbf{p}_{nr}[k] = \frac{1}{g_{cov}}\delta\mathbf{p}_{mrr}[k]$;

$$\begin{bmatrix}\delta u_{nr}[k]\\ \delta v_{nr}[k]\\ \delta w_{nr}[k]\end{bmatrix} = \text{VkT}\ \delta\mathbf{p}_{nr}[k];$$

$$\begin{bmatrix}\lambda_{\parallel m}[k]\\ \lambda_{\perp m}[k]\end{bmatrix} = (1-a_{cov})\begin{bmatrix}\text{lamdaTek1}\\ \text{lamdaNek1}\end{bmatrix} + a_{cov}\begin{bmatrix}\frac{\delta u_{nr}^2[k]+\delta v_{nr}^2[k]}{2}\\ \delta w_{nr}^2[k]\end{bmatrix};$$

$$\Delta\lambda_m[k] = \hat{\lambda}_{\parallel m}[k] - \hat{\lambda}_{\perp m}[k];$$

If $\frac{\Delta\lambda_m[k]}{\hat{\lambda}_{\parallel m}[k]} < \varepsilon$,

$$\begin{bmatrix}\theta_{xm}[k]\\ \theta_{ym}[k]\end{bmatrix} = \begin{bmatrix}\text{thetaxek1}\\ \text{thetayek1}\end{bmatrix};$$

Otherwise,

$$\begin{bmatrix}\theta_{xm}[k]\\ \theta_{ym}[k]\end{bmatrix} = \begin{bmatrix}\text{thetaxek1}\\ \text{thetayek1}\end{bmatrix} + a_{cov}\begin{bmatrix}\dfrac{\delta v_{nr}[k]\cdot\delta w_{nr}[k]}{\Delta\lambda_m[k]}\\ -\dfrac{\delta w_{nr}[k]\cdot\delta u_{nr}[k]}{\Delta\lambda_m[k]}\end{bmatrix}$$

$$\%\ {}^{V}\mathbf{y}_m[k] = \begin{bmatrix}{}^{V}\delta\mathbf{p}_m[k]\\ \boldsymbol{\lambda}_m[k]\\ \boldsymbol{\theta}_m[k]\end{bmatrix} = \begin{bmatrix}\delta u_m[k]\\ \delta v_m[k]\\ \delta w_m[k]\\ \lambda_{\parallel m}[k]\\ \lambda_{\perp m}[k]\\ \theta_{xm}[k]\\ \theta_{ym}[k]\end{bmatrix}$$

$$\mathbf{e}\text{pk} = \begin{bmatrix}\delta u_m[k]\\ \delta v_m[k]\\ \delta w_m[k]\end{bmatrix} - \text{Vdp1ek};$$

$$\mathbf{e}\text{lamdak} = \begin{bmatrix}\lambda_{\parallel m}[k]\\ \lambda_{\perp m}[k]\end{bmatrix} - \begin{bmatrix}\text{lamdaTek}\\ \text{lamdaNek}\end{bmatrix} = a_{cov}\left\{\begin{bmatrix}\frac{\delta u_{nr}^2[k]+\delta v_{nr}^2[k]}{2}\\ \delta w_{nr}^2[k]\end{bmatrix} - \begin{bmatrix}\text{lamdaTek}\\ \text{lamdaNek}\end{bmatrix}\right\};$$

$$\mathbf{e}\text{thetak} = \begin{bmatrix}\theta_{xm}[k]\\ \theta_{ym}[k]\end{bmatrix} - \begin{bmatrix}\text{thetaxek}\\ \text{thetayek}\end{bmatrix} = a_{cov}\begin{bmatrix}\dfrac{\delta v_{nr}[k]\cdot\delta w_{nr}[k]}{\Delta\lambda_m[k]}\\ -\dfrac{\delta w_{nr}[k]\cdot\delta u_{nr}[k]}{\Delta\lambda_m[k]}\end{bmatrix}$$

$$\mathbf{P}^f\mathbf{H}^T = \begin{bmatrix} \mathbf{P}_{11}^f & \mathbf{P}_{16}^f & \mathbf{P}_{18}^f \\ \mathbf{P}_{21}^f & \mathbf{P}_{26}^f & \mathbf{P}_{28}^f \\ \mathbf{P}_{31}^f & \mathbf{P}_{36}^f & \mathbf{P}_{38}^f \\ \mathbf{P}_{41}^f & \mathbf{P}_{46}^f & \mathbf{P}_{48}^f \\ \mathbf{P}_{51}^f & \mathbf{P}_{56}^f & \mathbf{P}_{58}^f \\ \mathbf{P}_{61}^f & \mathbf{P}_{66}^f & \mathbf{P}_{68}^f \\ \mathbf{P}_{71}^f & \mathbf{P}_{76}^f & \mathbf{P}_{78}^f \\ \mathbf{P}_{81}^f & \mathbf{P}_{86}^f & \mathbf{P}_{88}^f \\ \mathbf{P}_{91}^f & \mathbf{P}_{96}^f & \mathbf{P}_{98}^f \end{bmatrix}; \mathbf{H}\mathbf{P}^f\mathbf{H}^T = \begin{bmatrix} \mathbf{P}_{11}^f & \mathbf{P}_{16}^f & \mathbf{P}_{18}^f \\ \mathbf{P}_{61}^f & \mathbf{P}_{66}^f & \mathbf{P}_{68}^f \\ \mathbf{P}_{81}^f & \mathbf{P}_{86}^f & \mathbf{P}_{88}^f \end{bmatrix};$$

$\mathbf{R}[k] =?$

$$\mathbf{G} = \left\{ \begin{bmatrix} \mathbf{P}_{11}^f & \mathbf{P}_{16}^f & \mathbf{P}_{18}^f \\ \mathbf{P}_{61}^f & \mathbf{P}_{66}^f & \mathbf{P}_{68}^f \\ \mathbf{P}_{81}^f & \mathbf{P}_{86}^f & \mathbf{P}_{88}^f \end{bmatrix} + \mathbf{R}[k] \right\}^{-1};$$

$$\mathbf{L}[k] = \begin{bmatrix} \mathbf{L}_{11} & \mathbf{L}_{12} & \mathbf{L}_{13} \\ \mathbf{L}_{21} & \mathbf{L}_{22} & \mathbf{L}_{23} \\ \mathbf{L}_{31} & \mathbf{L}_{32} & \mathbf{L}_{33} \\ \mathbf{L}_{41} & \mathbf{L}_{42} & \mathbf{L}_{43} \\ \mathbf{L}_{51} & \mathbf{L}_{52} & \mathbf{L}_{53} \\ \mathbf{L}_{61} & \mathbf{L}_{62} & \mathbf{L}_{63} \\ \mathbf{L}_{71} & \mathbf{L}_{72} & \mathbf{L}_{73} \\ \mathbf{L}_{81} & \mathbf{L}_{82} & \mathbf{L}_{83} \\ \mathbf{L}_{91} & \mathbf{L}_{92} & \mathbf{L}_{93} \end{bmatrix} = \begin{bmatrix} \mathbf{P}_{11}^f & \mathbf{P}_{16}^f & \mathbf{P}_{18}^f \\ \mathbf{P}_{21}^f & \mathbf{P}_{26}^f & \mathbf{P}_{28}^f \\ \mathbf{P}_{31}^f & \mathbf{P}_{36}^f & \mathbf{P}_{38}^f \\ \mathbf{P}_{41}^f & \mathbf{P}_{46}^f & \mathbf{P}_{48}^f \\ \mathbf{P}_{51}^f & \mathbf{P}_{56}^f & \mathbf{P}_{58}^f \\ \mathbf{P}_{61}^f & \mathbf{P}_{66}^f & \mathbf{P}_{68}^f \\ \mathbf{P}_{71}^f & \mathbf{P}_{76}^f & \mathbf{P}_{78}^f \\ \mathbf{P}_{81}^f & \mathbf{P}_{86}^f & \mathbf{P}_{88}^f \\ \mathbf{P}_{91}^f & \mathbf{P}_{96}^f & \mathbf{P}_{98}^f \end{bmatrix} \mathbf{G};$$

$${}^V\delta\hat{\mathbf{p}}_1[k+1] = \mathrm{Vdp2ek} + \mathbf{L}11 * \mathbf{e}\mathrm{pk} + \mathbf{L}12 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}13 * \mathbf{e}\mathrm{thetak}$$

$${}^V\delta\hat{\mathbf{p}}_2[k+1] = \mathrm{Vdp3ek} + \mathbf{L}21 * \mathbf{e}\mathrm{pk} + \mathbf{L}22 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}23 * \mathbf{e}\mathrm{thetak}$$

$${}^V\delta\hat{\mathbf{p}}_3[k+1] = \lambda_c \cdot \mathrm{Vdp3ek} + \mathbf{L}31 * \mathbf{e}\mathrm{pk} + \mathbf{L}32 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}33 * \mathbf{e}\mathrm{thetak}$$

$${}^V\hat{\mathbf{d}}[k+1] = \mathrm{VDek} + \mathrm{VdDek} + \mathbf{L}41 * \mathbf{e}\mathrm{pk} + \mathbf{L}42 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}43 * \mathbf{e}\mathrm{thetak}$$

$${}^V\delta\hat{\mathbf{d}}[k+1] = \mathrm{VdDek} + \mathbf{L}51 * \mathbf{e}\mathrm{pk} + \mathbf{L}52 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}53 * \mathbf{e}\mathrm{thetak}$$

$$\hat{\boldsymbol{\lambda}}[\mathrm{k}+1] = \mathrm{lamdaek} + d\mathrm{lamdaek} + \mathbf{L}61 * \mathbf{e}\mathrm{pk} + \mathbf{L}62 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}63 * \mathbf{e}\mathrm{thetak}$$

$$\delta\hat{\boldsymbol{\lambda}}[\mathrm{k}+1] = d\mathrm{lamdaek} + \mathbf{L}71 * \mathbf{e}\mathrm{pk} + \mathbf{L}72 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}73 * \mathbf{e}\mathrm{thetak}$$

$$\widehat{\boldsymbol{\theta}}[\mathrm{k}+1] = \mathrm{thetaek} + d\mathrm{thetaek} + \mathbf{L}81 * \mathbf{e}\mathrm{pk} + \mathbf{L}82 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}83 * \mathbf{e}\mathrm{thetak}$$

$$\delta\widehat{\boldsymbol{\theta}}[\mathrm{k}+1] = d\mathrm{thetaek} + \mathbf{L}91 * \mathbf{e}\mathrm{pk} + \mathbf{L}92 * \mathbf{e}\mathrm{lamdak} + \mathbf{L}93 * \mathbf{e}\mathrm{thetak}$$

$$\delta\widehat{\mathbf{p}}_1[k+1] = \mathrm{dp1ek} = \mathrm{Vk} \cdot {}^V\delta\widehat{\mathbf{p}}_1[k+1];$$

$$\delta\widehat{\mathbf{p}}_2[k+1] = \mathrm{dp2ek} = \mathrm{Vk} \cdot {}^V\delta\widehat{\mathbf{p}}_2[k+1];$$

$$\delta\widehat{\mathbf{p}}_3[k+1] = \mathrm{dp3ek} = \mathrm{Vk} \cdot {}^V\delta\widehat{\mathbf{p}}_3[k+1];$$

$$\hat{\mathbf{d}}[k+1] = \mathrm{Dek} = \mathrm{Vk} \cdot {}^V\hat{\mathbf{d}}[k+1];\ \mathrm{dDek} = \mathrm{Vk} \cdot {}^V\delta\hat{\mathbf{d}}[k+1];$$

$$\Delta\hat{\lambda}[k] = \mathrm{lamdaTek} - \mathrm{lamdaNek}$$

$$\mathbf{G}_\lambda[k] = \begin{bmatrix} \Delta u[k] & 0 \\ \Delta v[k] & 0 \\ 0 & \Delta w[k] \end{bmatrix};$$

$$\mathbf{G}_\theta[k] = \begin{bmatrix} 0 & -\Delta\hat{\lambda}[k]\Delta w[k] \\ \Delta\hat{\lambda}[k]\Delta w[k] & 0 \\ \Delta\hat{\lambda}[k]\Delta v[k] & -\Delta\hat{\lambda}[k]\Delta u[k] \end{bmatrix};$$

$$\mathbf{F}[k] = \begin{bmatrix} 0_{3\mathrm{x}3} & \mathbf{I}_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} \\ 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & \mathbf{I}_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} \\ 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & \mathbf{I}_{3\mathrm{x}3} & -\mathbf{I}_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & -\mathbf{G}_\lambda[k] & 0 & -\mathbf{G}_\theta[k] & 0_{3\mathrm{x}2} \\ 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & \mathbf{I}_{3\mathrm{x}3} & \mathbf{I}_{3\mathrm{x}3} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} \\ 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & 0_{3\mathrm{x}3} & \mathbf{I}_{3\mathrm{x}3} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} & 0_{3\mathrm{x}2} \\ 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & \mathbf{I}_{2\mathrm{x}2} & \mathbf{I}_{2\mathrm{x}2} & 0_{2\mathrm{x}2} & 0_{2\mathrm{x}2} \\ 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}2} & \mathbf{I}_{2\mathrm{x}2} & 0_{2\mathrm{x}2} & 0_{2\mathrm{x}2} \\ 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}2} & 0_{2\mathrm{x}2} & \mathbf{I}_{2\mathrm{x}2} & \mathbf{I}_{2\mathrm{x}2} \\ 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}3} & 0_{2\mathrm{x}2} & 0_{2\mathrm{x}2} & 0_{2\mathrm{x}2} & \mathbf{I}_{2\mathrm{x}2} \end{bmatrix};$$

$$\mathrm{lamdaek} = \begin{bmatrix} \mathrm{lamdaTek} \\ \mathrm{lamdaNek} \end{bmatrix} = \widehat{\boldsymbol{\lambda}}[\mathrm{k}+1];\ d\mathrm{lamdaek} = \delta\widehat{\boldsymbol{\lambda}}[\mathrm{k}+1];$$

$$\mathrm{thetaek} = \begin{bmatrix} \mathrm{thetaxek} \\ \mathrm{thetayek} \end{bmatrix} = \widehat{\boldsymbol{\theta}}[\mathrm{k}+1];\ d\mathrm{thetaek} = \delta\widehat{\boldsymbol{\theta}}[\mathrm{k}+1]$$

$$c_x = \cos(\mathrm{thetaxek});\ s_x = \sin(\mathrm{thetaxek});\ c_y = \cos(\mathrm{thetayek});\ s_y = \sin(\mathrm{thetayek});$$

$$\widehat{\mathbf{V}}[k+1] = \mathrm{Vk} = \begin{bmatrix} c_y & s_y s_x & s_y c_x \\ 0 & c_x & -s_x \\ -s_y & c_y s_x & c_y c_x \end{bmatrix};\ \mathrm{VkT} = \begin{bmatrix} c_y & 0 & -s_y \\ s_y s_x & c_x & c_y s_x \\ s_y c_x & -s_x & c_y c_x \end{bmatrix}$$

$\mathrm{Vdp1ek} = \mathrm{VkT} \cdot \mathrm{dp1ek}$; $\mathrm{Vdp2ek} = \mathrm{VkT} \cdot \mathrm{dp2ek}$; $\mathrm{Vdp3ek} = \mathrm{VkT} \cdot \mathrm{dp3ek}$;

$\mathrm{VDek} = \mathrm{VkT} \cdot \mathrm{Dek}$; $\mathrm{VdDek} = \mathrm{VkT} \cdot \mathrm{dDek}$;

$\mathbf{P} = (\mathbf{I} - \mathbf{L}[k]\mathbf{H})\mathbf{P}^f$;

$\mathbf{P}^f = \mathbf{F}[k]\mathbf{P}\mathbf{F}^T[k] + \mathbf{Q}[k]$; % $\mathbf{P}k$ is the error covariance and $\mathbf{Q}[k]$ is the variance of the process noise.

$\mathbf{Q}_{11}[k] = \mathbf{0}$

$\mathbf{Q}_{22}[k] = \mathbf{0}$

$$\mathbf{Q}_{33}[k] = 4k_B T\left(\frac{\Delta t}{\gamma_N}\right)\widehat{\mathbf{\Lambda}}[k] = (0.01589)^2\begin{bmatrix} \hat{\lambda}_{\parallel}[k] & 0 & 0 \\ 0 & \hat{\lambda}_{\parallel}[k] & 0 \\ 0 & 0 & \hat{\lambda}_{\perp}[k] \end{bmatrix}$$

$\mathbf{Q}_{44}[k] = a_{pd} \cdot a_{prd} \cdot \mathbf{Q}_{33}[k]$

$\mathbf{Q}_{55}[k] = a_{pd} \cdot a_{prd} \cdot \mathbf{Q}_{33}[k]$

$$\mathbf{Q}_{66}[k] = a_{cov} \cdot (0.01589)^2\begin{bmatrix} \hat{\lambda}_{\parallel}[k] & 0 \\ 0 & \hat{\lambda}_{\perp}[k] \end{bmatrix}$$

$\mathbf{Q}_{77}[k] = \mathbf{Q}_{66}[k]$

$\mathbf{Q}_{88}[k1] =?$

$\mathbf{Q}_{99}[k1] =?$